

HELPDESK CENTER — PROJECT CHARTER

EXECUTIVE SUMMARY

Helpdesk Center is an AI-assisted internal helpdesk ticketing system designed to automate issue categorization and streamline support workflows. The system enables employees to submit support tickets with attachments while IT and HR agents manage resolution through role-based dashboards.

IBM Watson Natural Language Understanding automatically categorizes tickets into Hardware, Software, or HR/Facilities departments. Priority assignment is automated through keyword analysis. The application is built using Spring Boot and React, deployed on IBM Cloud Code Engine with PostgreSQL database and IBM Cloud Object Storage for file management.

This project demonstrates full-stack development proficiency, cloud-native architecture, AI integration, and enterprise system design capabilities required for capstone completion.

PROJECT BACKGROUND

1. Problem Statement

Organizations currently manage internal support requests through fragmented channels (email, chat, phone), resulting in:

- Manual ticket routing and categorization
- Delayed response times and lost requests
- Lack of visibility and accountability
- Inefficient resource allocation

2. Proposed Solution

An automated ticketing system that centralizes request intake, leverages AI for categorization, and provides role-based access for support teams, thereby reducing manual overhead and improving resolution efficiency.

PROJECT OBJECTIVES

1. Develop a web-based ticketing system with user authentication and role-based access control
2. Implement AI-powered automatic ticket categorization using IBM Watson NLU
3. Enable keyword-based automatic priority assignment
4. Provide role-specific dashboards for IT Hardware, IT Software, and HR support agents
5. Integrate secure file attachment functionality using IBM Cloud Object Storage
6. Deploy the complete system on IBM Cloud Code Engine

PROJECT SCOPE

1. In-scope Features

- User authentication with role-based access (Employee, IT Hardware, IT Software, HR)
- Ticket submission with title, description, email, and file attachments (maximum 3 files, 10MB each)
- AI-powered categorization (Hardware, Software, HR/Facilities)
- Automated priority assignment (Low, Medium, High, Critical)
- Ticket lifecycle management (Open, In Progress, Resolved)
- Role-filtered dashboards
- Comment/update system for agent collaboration
- File upload to IBM Cloud Object Storage with presigned URL generation
- Filtering and sorting capabilities (by category, status, priority, date)

2. Out-of-scope Features

- Automated email notifications
- Real-time chat functionality
- Mobile application
- SLA tracking and escalation workflows
- Analytics dashboard
- Multi-language support
- User management administrative panel
- Integration with external systems (Slack, Teams, Jira)

DELIVERABLES

1. Technical Deliverables

- Fully functional web application (Spring Boot backend, React frontend)
- PostgreSQL database with complete schema
- Deployed system on IBM Cloud Code Engine
- Docker container image
- Source code repository

2. Documentation Deliverables

- Project Charter (this document)
- System Architecture Documentation
- Entity Relationship Diagram (ERD)
- API Documentation
- Deployment Guide
- User Manual
- Capstone Report

STAKEHOLDERS

Role	Responsibility	Interest Level
Student Developer	System design, implementation, deployment	High
Capstone Adviser	Project guidance and evaluation	High
Academic Panel	Project assessment	High
End Users (Employees)	Ticket submission	Medium
Support Agents (IT/HR)	Ticket resolution	Medium

FUNCTIONAL REQUIREMENTS

1. Authentication & Authorization

- Secure login with username and password (BCrypt hashing)
- Role-based access control (Employee, IT Hardware, IT Software, HR)
- Session management

2. Ticket Management

- Create tickets with mandatory fields (title, description, email)
- Optional file attachments (1-3 files)
- Automatic AI categorization via Watson NLU
- Automatic priority assignment based on keyword analysis
- Status workflow: Open → In Progress → Resolved
- View, filter, and sort ticket lists

3. Agent Operations

- Role-specific ticket dashboards
- Add comments and updates
- Change ticket status
- Download attachments
- Mark tickets as resolved

4. File Management

- Upload files to IBM Cloud Object Storage
- Generate presigned URLs for secure upload
- Store file metadata in database
- Support common file formats (images, PDFs, documents)

NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Target
Performance	Dashboard load time	< 2 seconds
Performance	Ticket submission time	< 3 seconds
Performance	Concurrent users	100 users
Security	Password encryption	BCrypt
Security	Input validation	SQL injection & XSS prevention
Security	Data transmission	HTTPS in production
Reliability	System uptime	99%
Reliability	AI failure handling	Graceful fallback
Scalability	Maximum tickets	1,000 tickets

Scalability	Maximum users	200 users
Usability	User training required	None
Compatibility	Browser support	Chrome, Firefox, Safari, Edge

SYSTEM ARCHITECTURE

Multi-Tier Application Architecture Diagram

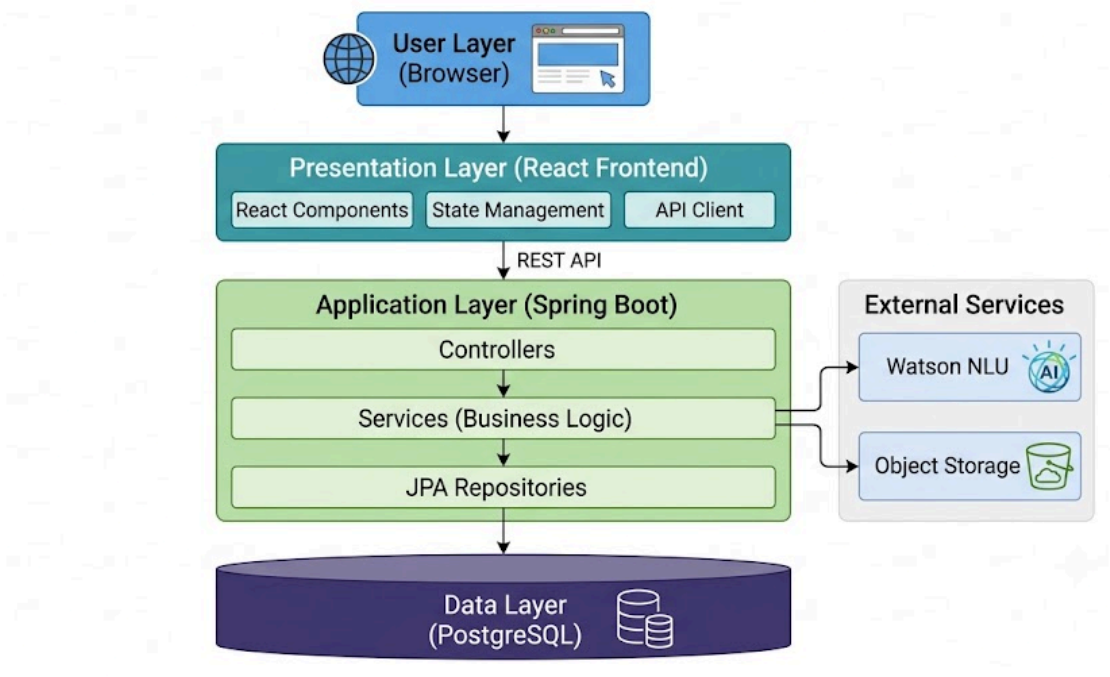


Figure 1. Helpdesk Center System Architecture

TECHNOLOGY STACK

Component	Technology	Version
Backend Framework	Spring Boot	3.2.x
Frontend Framework	React	18.x
Programming Language	Java	17

Database	PostgreSQL	15.x
ORM	Spring Data JPA	3.2.x
AI Service	IBM Watson NLU	Cloud
File Storage	IBM Cloud Object Storage	Cloud
Deployment Platform	IBM Cloud Code Engine	Cloud
Containerization	Docker	Latest
Build Tool	Maven	3.9.x

DATA MODEL

1. Core Entities

Users

- id, name, email, password_hash, role, created_at

Tickets

- id, title, description, email, category, priority, status, created_at, updated_at, created_by (FK), assigned_to (FK)

Comments

- id, ticket_id (FK), user_id (FK), message, created_at

Attachments

- id, ticket_id (FK), file_name, file_url, file_size, uploaded_at

2. Relationships

Users → Tickets (1:M)

Tickets → Comments (1:M)

Tickets → Attachments (1:M)

Users → Comments (1:M)

PROJECT TIMELINE

Phase	Key Activities	Deliverable
Day 1	Backend setup, database schema, authentication, basic CRUD	Working backend API

Day 2	AI integration (Watson NLU), priority logic, file upload (Object Storage)	Auto-categorization functional
Day 3	React frontend development, API integration	Complete UI
Day 4	Comment system, filtering, sorting, bug fixes	Fully functional system
Day 5	Docker containerization, deployment, documentation, presentation prep	Deployed system + documentation

RISK MANAGEMENT

Risk	Impact	Probability	Mitigation Strategy
Watson NLU API failure	High	Low	Implement fallback to "General Support" category
Time constraint	High	Medium	Strict scope control, no feature additions
File upload complexity	Medium	Medium	Use presigned URLs, limit file size to 10MB
Integration issues	Medium	Medium	Test APIs independently before integration
Deployment failures	Medium	Low	Local Docker testing before cloud deployment

ASSUMPTIONS

- IBM Cloud free tier resources are available and sufficient
- Watson NLU Lite plan supports required API call volume
- Users have stable internet connectivity
- Manual email communication by agents is acceptable (no automated notifications)
- English language only for AI classification

- System operates within MVP scale (≤ 200 users, ≤ 1000 tickets)

SUCCESS CRITERIA

The project will be considered successful upon meeting the following criteria:

- Functional ticket submission and retrieval system
- AI categorization accuracy $\geq 70\%$
- Automated priority assignment based on keywords
- Role-based dashboards operational
- File upload and download functionality working
- System deployed and accessible via IBM Cloud
- Complete documentation delivered
- Successful capstone defense presentation